

On the power-series construction of bound states. I. The energies as zeros of the infinite Hill determinants

M. Znojil

Institute of Nuclear Physics, Czechoslovak Academy of Sciences, 250 68 Řež, Czechoslovakia

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For a broad class of potentials, we show that the nonvariational (so-called Hill-determinant) intuitive identification of binding energies with zeros of certain infinite determinants may be given a rigorous mathematical foundation. The essence of the construction lies in an appropriate restriction of the class of the admissible *Ansätze*. This eliminates the undesirable confluence of the physical and unphysical components of $\psi(r)$ that may take place in the general case.

I. INTRODUCTION

An exceptional simplicity of the harmonic oscillator wave functions

$$\psi_n(r) = \exp\left(-\frac{1}{2}r^2\right) \sum_{m=0}^n p_m r^{2m+l+1} \quad (1.1)$$

(cf., e.g., Flügge¹) forms a natural background for the perturbative² as well as variational³ treatment of anharmonicities. Recently, the direct generalizations of (1.1),

$$\psi_n(r) = \exp(\text{polynomial}) \times \text{power series}, \quad (1.2)$$

were also studied in the numerical context,^{4,5} with the intention of deriving the new resummation techniques,⁶ etc. Besides the simplicity and semianalytic character of the wave functions, additional merit of (1.2) lies also in the possibility of an approximative replacement of the power series by a polynomial, and in the related tractability of the approximate energies as zeros of the so-called Hill determinants.⁷

In a purely formal manner, the "generalized harmonic oscillator" wave functions (1.2) may be interpreted as an *Ansatz*, converting, say, the radial differential Schrödinger equation

$$\left[-\frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} + V(r)\right]\psi(r) = E\psi(r) \quad (1.3)$$

into a purely algebraic problem.^{4,7,8} Unfortunately, in contrast to the more standard numerical and variational methods,³ it is not always easy to treat properly also the physical boundary conditions in the new algebraic language.⁴ Even the simplest example with the special sextic anharmonic oscillator potential,

$$V(r) = ar^2 + br^4 + cr^6, \quad c > 0, \quad b < 0,$$

was not treated properly from this point of view.^{6,9} This inspired an extensive discussion in the literature (cf., e.g., Ref. 10 for the review).

At present, the latter $b < 0$ sextic oscillator puzzle seems resolved. Singh's "completely WKB" choice of the polynomial exponent $P(r)$ in (1.2) has been proved incompatible¹¹ with the intuitive identification of the physical bound-state energies with zeros of the related "Hill determinants."⁷ Conversely, rigorous validity of the method has been confirmed for $b > 0$,¹² or for certain incompletely WKB¹³ or completely non-WKB⁵ modified exponents $P(r)$ in (1.2).

For a broader class of potentials, the situation will be clarified in the present paper. In Sec. II we specify the forces in more detail, in Sec. III we characterize them formally as a natural generalization of the harmonic oscillator, and, separating their class into a pair of its natural subclasses, we deliver the corresponding construction and proofs in Secs. IV and V. Section VI is a summary.

II. THE CLASS OF FORCES

An interest in simple potentials stems from the phenomenological needs of the atomic and molecular physics as well as from the methodical considerations of the perturbative field theory.¹⁴ In our preceding paper,¹⁵ our interest has been concentrated upon the general superpositions of the rational powers of the coordinate

$$V_0(r) = \sum_{i=1}^N g_i^{(0)} r^{m_i/n_i}, \quad m_N/n_N > \cdots > m_1/n_1 > -2. \quad (2.1)$$

These forces were shown to be tractable by the Hill-determinant technique, provided only that a certain "superconfinement restriction" (schematically, $g_N^{(0)} > 0$) was satisfied by the coupling constants. Now we intend to get rid of it. Indeed, a formulation of a procedure independent of the particular coupling values is important methodically. Moreover, our choice and study of the class of forces (2.1) may also be shown useful from the following points of view.¹⁵

(a) Their flexibility forms a good background for approximations of the various realistic forces. If needed, a component $g_0^{(0)} r^{-2}$ may also be used and incorporated into the centrifugal term, $l(l+1) + g_0^{(0)} = l'(l'+1)$. The condition of regularity $l' > -\frac{1}{2}$ implies that the coupling $g_0^{(0)} > -(l + \frac{1}{2})^2$ cannot be arbitrary.

(b) The restrictions imposed upon the exponents m_i/n_i and the singular coupling $g_0^{(0)}$ are minimal. They merely reflect an acceptable nonregularity of the differential equation and of its general solution near the origin,

$$\psi(r) \sim c_1 r^{l'+1} + c_2 r^{-l'}, \quad r \approx 0.$$

The latter, irregular component of $\psi(r)$ may immediately be omitted as unphysical.¹⁶

(c) Asymptotically, formula (2.1) incorporates both the finite wells ($m_N/n_N < 0$) and the confining potentials

($g_N^{(0)} > 0$, $m_N/n_N > 0$, and $E > E_{\min}$). In the wave functions, an asymptotic appearance of a node, $\psi(\infty) = 0$, is in a one-to-one correspondence to an emergence of a new bound state. This is valid within the discrete spectrum domain (for $E < g_N^{(0)}$ if $m_N/n_N = 0$) and follows from the standard Sturm-Liouville oscillation theorems.¹⁷ In particular, we shall have also $|\psi(r)| \sim \exp(\text{polynomial}) \gg 1$, for large $r \gg 1$ and for $E \neq E_{\text{phys}}$.

(d) An elementary change of variables,

$$\psi \rightarrow \psi_i, \quad r \rightarrow r_i, \quad l \rightarrow l_i, \quad \{E, g_n\} \rightarrow \{g_n^{(i)}, E_i\}, \quad 1 \leq i \leq i_{\max},$$

in Eq. (1.3) may be used to preserve its Schrödinger equation form and modify the potential¹⁸ $V_0 \rightarrow V_i$. In particular, we may obtain the even polynomials

$$V_1(r^2) = \sum_{m=0}^s g_m^{(1)} r^{2m}, \quad g_s > 0, \quad (2.2a)$$

ordinary polynomials

$$V_2(r) = \sum_{n=-1}^t g_n^{(2)} r^n, \quad g_t > 0 \text{ for } t \geq 1, \quad (2.2b)$$

and all the fractional-power superpositions up to $i = i_{\max} = I$ and $V_I(\infty) = \text{const} < \infty$,

$$V_I(r) = \sum_{k=0}^T g_k^{(I)} r^{-2k/T}. \quad (2.3)$$

All these potentials are equivalent, formally, to the original force $V_0 = V_i$, and we may study any of them without loss of generality. In the paper, we shall use only the polynomial forms (2.2), recalling the special case of their mutual transformation.¹⁸

$$\begin{aligned} \psi_2(r_2) &= \sqrt{r_1} \psi_1(r_1), \quad r_2 = r_1^2, \quad 2l_2 + 1 = l_1 + \frac{1}{2}, \\ g_{-1}^{(2)} &= \frac{1}{4}(g_0^{(1)} - E_1), \quad g_0^{(2)} - E_2 = \frac{1}{4}g_1^{(1)}, \\ g_m^{(2)} &= \frac{1}{4}g_{m+1}^{(1)}, \quad m = 1, 2, \dots, \quad s-1 \equiv t, \end{aligned} \quad (2.4)$$

as an illustration of the above implicit formula.

An analysis of the polynomial potentials (2.2) leads to an important formal difference between the even and odd degree t . Indeed, the general asymptotic solution of Eq. (1.3) reads

$$\psi(r) \approx d_1 \exp\left(\frac{ar^{t/2+1}}{t/2+1}\right) + d_2 \exp\left(\frac{-ar^{t/2+1}}{t/2+1}\right), \quad r \gg 1, \\ a^2 = g_t^{(2)}, \quad t \geq 1, \quad (2.5)$$

$$a^2 = g_0^{(2)} - E, \quad t < 1,$$

and may be used as a factor in Eq. (1.2) for $t = 2q$ only. In Ref. 15, an elimination of the square-root variable $r^{1/2}$ has been achieved by means of a transition (2.4) from odd t and $V = V_2$ to its even-degree representation V_1 with $t^{\text{new}} = 4q$, $t^{\text{old}} = 2q - 1$. Here, we shall treat both parities of t separately.

III. HARMONIC OSCILLATOR AS A METHODOLOGICAL GUIDE

The simplest form of *Ansatz* (1.2) reads

$$\psi(r) = e^{-\beta r^2/2} \sum_{n=0}^N p_n r^{2n+l+1}, \quad N \rightarrow \infty, \quad (3.1)$$

and may be inserted in our Schrödinger equation (1.3) with an arbitrary potential in principle. In order to illustrate this procedure, let us contemplate the simplest potential $V(r) = a^2 r^2$. Obviously, the recurrences obtained from (1.3),

$$\begin{aligned} B_n p_{n+1} &= (A_n - E) p_n + C_n p_{n-1}, \quad n = 0, 1, \dots, \\ B_n &= (2n+2)(2n+2l+3), \\ A_n &= (4n+2l+3)\beta, \quad C_n = a^2 - \beta^2, \end{aligned} \quad (3.2)$$

may immediately be solved by means of the explicit determinantal formula

$$p_{n+1} = \left(\prod_{k=1}^n B_k \right)^{-1} \cdot p_0 \det \begin{pmatrix} A_0 - E, & -B_0, & 0, \dots \\ 0, \dots, 0, & C_k, & A_k - E, & -B_k, & 0, \dots \\ 0, \dots, & 0, & C_n, & A_n - E, \end{pmatrix}, \quad n = 0, 1, \dots, \quad (3.3)$$

for an arbitrary choice of the parameter β in (3.1).

Of course, in the particular example in question, a "clever" choice of the parameter $\beta = a$ implies that $C_k = 0$ for all k . This makes the regular infinite series (1.2) coincide with the confluent hypergeometric function. It will terminate at the exceptional values $E = E_n = (4n+2l+3)a$, $n = 0, 1, \dots$, of energies. Then the overall exponential factor makes the norm of $\psi(r)$ finite—the exceptional energies form a complete spectrum. Formally, we may write a determinantal equation that defines exactly the first $N-1$ binding energies,⁷

$$p_N = 0, \quad N \gg 1. \quad (3.4)$$

The oversimplified harmonic oscillator example may

simulate the more complicated cases, provided that we pick up a "wrong" value of the parameter $\beta (\neq a)$ in (3.1). Then, in a purely heuristic manner, we may still simplify Eq. (3.2) (three-term recurrences) by a change of variables:

$$p_n = \beta^n q_n / n!, \quad n \geq 0. \quad (3.5)$$

In an asymptotic domain of indices, a simplified form of (3.2),

$$q_{n+1} - q_n = \frac{1}{4}(a^2/\beta^2 - 1)q_{n-1}, \quad n \gg 1, \quad (3.6)$$

may be used in place of the original difference equation.

The new equation has constant coefficients and may be solved by the *Ansatz*

$$q_n = \lambda^n.$$

After a determination of $\lambda = \lambda_{\pm} = (1 \pm a/\beta)/2$ from (3.6), the general solution of Eq. (3.2) becomes available,

$$p_n = \frac{c_1}{n!} \left(\frac{\beta + a}{2} \right)^n + \frac{c_2}{n!} \left(\frac{\beta - a}{2} \right)^n, \quad n \gg 1, \quad (3.7)$$

and leads also to the $\beta = a$ termination condition. Of course, its relation to the normalizability physical requirement becomes less obvious.

In the detailed analysis, we may recall (2.5), i.e.,

$$\psi(r) = d_1 e^{ar^2/2} + d_2 e^{-ar^2/2}, \quad r \gg 1, \quad (3.8)$$

where $d_1 = 0$ if and only if $E = E_{\text{phys}}$.

At the same time, we may insert (3.7) in (3.1) and write

$$\begin{aligned} \psi(r) \exp(\beta r^2) &\approx \text{polynomial} + \sum_{n \gg N} \dots \\ &\approx c_1 \exp\left(\frac{\beta + a}{2} r^2\right) + c_2 \exp\left(\frac{\beta - a}{2} r^2\right) \\ &\quad + O(r^{2N}), \quad r \gg 1, \quad N \gg 1. \end{aligned} \quad (3.9)$$

Thus, provided that $\beta > 0$, the asymptotics of $\psi(r)$, $r \rightarrow \infty$, remain unphysical ($d_1 \neq 0$) whenever $c_1 \neq 0$ in Eq. (3.7) or (3.9). Conversely, both the coefficients d_1 and c_1 change sign precisely at $E = E_{\text{phys}}$ when treated as functions of the variable E . Obviously, we may write

$$\begin{aligned} N! p_N &= \left(\frac{\beta + a}{2} \right)^N \left(c_1 + c_2 \left(\frac{\beta - a}{\beta + a} \right)^N \right) \\ &= c_1 ((\beta + a)/2)^N (1 + O(\epsilon^N)), \\ 0 < \epsilon &= (\beta - a)/(\beta + a) < 1, \end{aligned} \quad (3.10)$$

and see that the roots of p_N coincide with the physical binding energies E_n in the limit $N \rightarrow \infty$. In this way, a rigorous foundation of the Hill-determinant method is obtained—in accord with (3.3), we may employ the approximate secular equation (3.4),

$$\det \left[\begin{pmatrix} A_0 & -B_0 & & \\ C_1 & A_1 & -B_1 & \\ & \ddots & \ddots & \\ & & C_N & A_N \end{pmatrix} - EI \right] = 0, \quad N \gg 1, \quad (3.11)$$

and determine the binding energies by the standard computation algorithms.³

$$p_{n+1} = \left(\prod_{k=0}^n B_k \right)^{-1} p_0 \det \begin{pmatrix} A_0 - E, & -B_0, & 0, \dots \\ C_1^{(1)}, & A_1 - E, & -B_1, & 0, \dots \\ & \ddots & \ddots & \\ 0, \dots, 0, & D^{(q+1)}, \dots, D^{(1)}, & C_n^{(q-1)}, \dots, C_n^{(1)}, & A_n - E \end{pmatrix}. \quad (4.6)$$

Hence the standard finite-dimensional analogies make no sense in the infinite-dimensional system in question.¹⁰ A detailed analysis of relations between the physical asymptotics of $\psi(r)$, $r \rightarrow \infty$, and p_n , $n \rightarrow \infty$, necessitates a more careful argumentation as indicated schematically in the preceding section.

IV. POTENTIALS EQUIVALENT TO POLYNOMIALS OF AN ODD DEGREE

For $t = 2q - 1$ in the definition of the potentials $V = V_2(r)$, we may use the change of variables (2.4) and consider an equivalent potential

$$V(r) = g_0 + g_1 r^2 + \dots + g_{2q} r^{4q}, \quad g_{2q} = a^2 > 0, \quad (4.1)$$

giving the quartic anharmonic oscillator in the simplest non-trivial $q = 1$ case. In accord with the standard Hill-determinant computations,⁷ we may choose the exponential (subdominant) factor in an almost arbitrary way in (1.2),

$$\psi(r) = r^{l+1} \exp(-g(r)) \sum_{n=0}^{\infty} p_n r^{2n}, \quad (4.2)$$

$$g(r) = \sum_{j=1}^q \beta_j r^{2j}/2j.$$

Then, an insertion in Eqs. (1.3) and (4.1) leads to the basic $(2q + 2)$ -term recurrences of the type (3.2),

$$\begin{aligned} B_n p_{n+1} &= (A_n - E) p_n + \sum_{j=1}^{q-1} p_{n-j} C_n^{(j)} \\ &\quad + \sum_{j=q}^{2q} p_{n-j} D^{(j+1-q)}, \\ n &= 0, 1, \dots, \quad p_{-1} = p_{-2} = \dots = 0. \end{aligned} \quad (4.3)$$

Of course, the coefficients are functions of the couplings and other parameters,

$$\begin{aligned} B_n &= (2n + 2)(2n + 2l + 3), \\ A_n &= (4n + 2l + 3)\beta_1 + g_0 \equiv C_n^{(0)} + E, \\ C_n^{(j)} &= (4n + 2l + 3 - 2j)\beta_{j+1} + g_j \\ &\quad + \sum_{i=0}^{j-1} \beta_{i+1} \beta_{j-i}, \\ j &= 1, 2, \dots, q-1, \quad n = 0, 1, \dots, \\ D^{(j)} &= g_{j+q-1} + \sum_{i=j-1}^{q-1} \beta_{i+1} \beta_{q+j-i-1}, \\ j &= 1, 2, \dots, q, \\ D^{(q+1)} &= g_{2q} = a^2. \end{aligned} \quad (4.4)$$

A discussion of the physical normalization requirement remains an open question here.

In general, we may expect an asymptotic growth of $\psi(r)$:

$$\psi(r) \approx d_1 \exp\left(\frac{ar^{2q+1}}{2q+1}\right) + d_2 \exp\left(\frac{-ar^{2q+1}}{2q+1}\right), \quad r \gg 1. \quad (4.5)$$

At the same time, each row of Eq. (4.3) defines the nonzero coefficient p_{n+1} at each energy E ,

In the first step, we shall employ a change of variables inspired by Eq. (3.5),

$$p_{n+1} = \frac{\lambda^n h_n}{\Gamma(1 + (n+1)/(2q+1))\Gamma(1 + (n+l+\frac{3}{2})/(2q+1))}. \quad (4.7)$$

This converts our basic recurrences (4.3) into a set of the alternative difference equations

$$h_n - h_{n-2q-1} = \sum_{m=0}^{q-1} (w_{q-m} h_{n+m-q} + w_{2q-m} h_{n+m-2q}), \quad (4.8)$$

each of which may be numbered by the parameter k such that

$$\lambda = \lambda_{(k)} = \left[\frac{a^2}{(4q+2)^2} \right]^{1/(2q+1)} \exp\left(i \frac{2\pi k}{2q+1}\right), \quad k = 0, \pm 1, \pm 2, \dots, \pm q, \quad (4.9)$$

in the *Ansatz* (4.7). The new values of coefficients in (4.8),

$$\begin{aligned} w_{q-m} &= \frac{C_n^{(q-1-m)} \Gamma(1 + (n+1)/(2q+1)) \Gamma(1 + (n+l+\frac{3}{2})/(2q+1))}{\lambda^{q-m} B_n \Gamma(1 + (n+1+m-q)/(2q+1)) \Gamma(1 + (n+l+\frac{3}{2}+m-q)/(2q+1))}, \\ w_{2q-m} &= \frac{D^{(q-m)} \Gamma(1 + (n+1)/(2q+1)) \Gamma(1 + (n+l+\frac{3}{2})/(2q+1))}{\lambda^{2q-m} B_n \Gamma(1 + (1+n+m-2q)/(2q+1)) \Gamma(1 + (n+l+\frac{3}{2}+m-2q)/(2q+1))}, \\ m &= 0, 1, \dots, q-1, \end{aligned} \quad (4.10)$$

are ordered with respect to their asymptotic decrease [cf. Fig. 1(a)],

$$\begin{aligned} w_{q-m} &= O(n^{-(2m+1)/(2q+1)}), \\ w_{2q-m} &= O(n^{-(2m+2)/(2q+1)}). \end{aligned} \quad (4.11)$$

In comparison with the harmonic oscillator methodical example (3.6), our equations (4.8) have an asymptotically negligible right-hand-side expression—one of their solutions should obviously be almost constant for $n \gg 1$.

The almost-constant solution may easily be represented by its Taylor series truncated after a few terms. In this case, we may replace also each of the difference equations (4.8) by the simplest nontrivial differential equation approximation,

$$(2q+1) \frac{d}{dn} h_n = w_q h_n + \text{corrections}, \quad n \gg 1, \quad (4.12)$$

and derive immediately the leading-order form of the almost-constant solution. Assuming for simplicity that $\beta_q \neq 0$, we get

$$\begin{aligned} w_q &= (2q+1)^{-1} \beta_q [n/(2q+1)]^{-1/(2q+1)} / \lambda_{(k)}^q \\ &+ \text{corrections}, \end{aligned}$$

i.e.,

$$\begin{aligned} h_n = h_n^{(k)} &= \exp \left\{ \frac{\beta_q}{2q\lambda_{(k)}^q} [(2q+1)^{-1}n]^{2q/(2q+1)} \right. \\ &\left. + \text{corrections} \right\}, \quad n \gg 1, \end{aligned} \quad (4.13)$$

for each particular parameter k . This is similar to the construction of the harmonic oscillator solutions for each $e = e_{\pm}$. The general solution of the present difference equation (4.8) may be written as a superposition of the type (3.7),

$$\begin{aligned} &\Gamma\left(1 + \frac{n+1}{2q+1}\right) \Gamma\left(1 + \frac{n+l+\frac{3}{2}}{2q+1}\right) p_{n+1} \\ &= \sum_{k=-q}^q c_k \lambda_{(k)}^n h_n^{(k)}, \quad n \gg 1. \end{aligned} \quad (4.14)$$

This completes the first part of our analysis—an insertion of (4.13) will enable us to derive the present analog of the harmonic oscillator formula (3.10).

The proof of the previous statement is not complicated. First, we renumber the roots (4.9) in such a way that $\lambda_{(k)}^q = \lambda_{(k_0)} / \lambda_0^{q+1}$, i.e., we put $k_0 = j - q$ for $k = 2j + 1$, $k_0 = j$

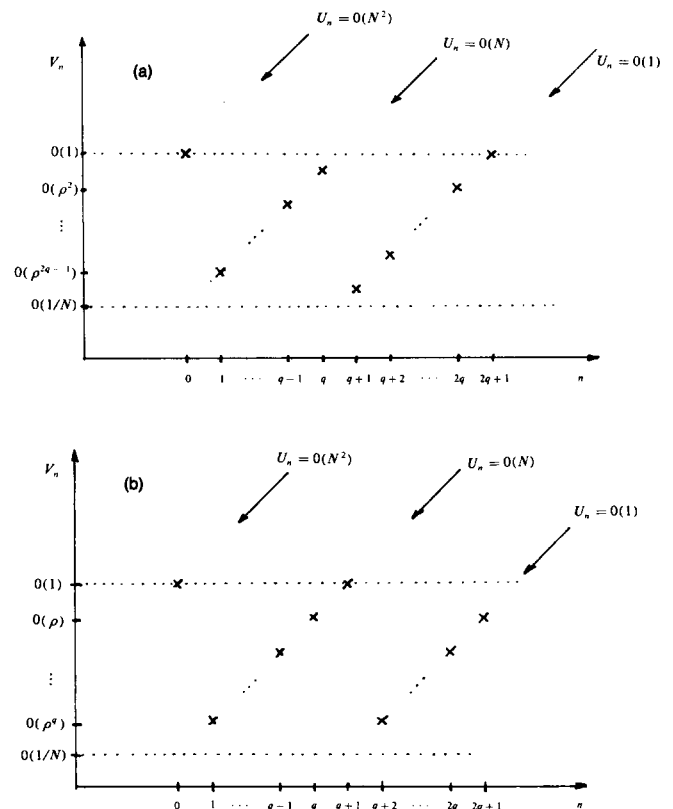


FIG. 1. Order of magnitude of coefficients in our recurrences

$\sum_{j=0}^{2q+1} U_j p_{N+1-j} = 0$ and $\sum_{j=0}^{2q+1} V_j h_{N-j} = 0$,
for large $N \gg 1$: (a) $t = 2q - 1$, $\rho = N^{-1/(2q+1)}$; (b) $t = 2q$, $\rho = N^{-1/(q+1)}$.

for $k = 2j$, and $k_0 = 0$ if and only if $k = 0$. Next, we notice that $\text{Re } \lambda_{(k_0)} < \lambda_{(0)}$ for all $k_0 \neq 0$. Thus due to (4.13) we have $|h_n^{(k)}| \ll |h_n^{(0)}|$, for all $k \neq 0$ and $n \gg 1$. This means that in full analogy to (3.10), we may neglect all the exponentially small corrections and write, for sufficiently large indices N ,

$$\begin{aligned} & \Gamma\left(1 + \frac{N+1}{2q+1}\right) \Gamma\left(1 + \frac{N+l+\frac{3}{2}}{2q+1}\right) p_{N+1} \\ & \approx c_0 \lambda_{(0)}^{N_{(0)}} h_N^{(0)} \\ & \approx c_0 \lambda_{(0)}^{N_{(0)}} \exp\left\{\frac{\beta_q}{2q\lambda_{(0)}^q} \left[\frac{N}{2q+1}\right]^{2q/(2q+1)}\right\}, \quad N \gg 1. \end{aligned} \quad (4.15)$$

The asymptotic zeros of p_N and c_0 will coincide.

In the next step of our considerations, we may try to extend the asymptotic estimate (3.9) beyond the harmonic oscillator trivial case. Thus combining (4.2) with (4.15) we arrive at an estimate

$$\psi(r) = r^{l+1} \exp(-g(r)) \left(O(r^{2N}) + \sum_{h=N}^{\infty} p_h r^{2h} \right), \quad (4.16)$$

where the summation may be approximated by an integration,

$$\phi_N(r) = \sum_{n=N}^{\infty} p_n r^{2n} \approx c_0 \int_N^{\infty} dn \frac{r^{2n} \exp(n \ln \lambda_{(0)} + \text{corrections})}{\Gamma(1 + (n+1)/(2q+1)) \Gamma(1 + (n+l+\frac{3}{2})/(2q+1))}, \quad N \gg 1. \quad (4.17)$$

Then a change of variables leads to the formula

$$\begin{aligned} \phi_N(r) & \approx \frac{2q+1}{2} c_0 \int_{N/(q+1/2)}^{\infty} dm \frac{r^{(2q+1)m} \exp[(q+\frac{1}{2})m \ln \lambda_{(0)} + \text{corrections}]}{\Gamma(1 + m/2 + 1/(2q+1)) \Gamma(1 + m/2 + (l+\frac{3}{2})/(2q+1))} \\ & \approx c_0 \sum_{m=\{N/(q+\frac{1}{2})\}}^{\infty} \frac{r^{m(2q+1)}}{m!} \exp\left[m\left(\ln 2 + \frac{2q+1}{2} \ln \lambda_0\right) (1 + O(m^{-1/(2q+1)}))\right] \\ & \approx c_0 \exp[ar^{2q+1}/(2q+1) + \text{corrections}], \quad r \gg 1, \end{aligned} \quad (4.18)$$

compatible, within the achieved precision, with the asymptotic $d_1 \neq 0$ estimate (4.5).

A common zero in c_0 , $\phi_N(r)$, or $\psi(r)$ is in a one-to-one correspondence to the energy, crossing its bound-state value $E = E_{\text{phys}}$. We may conclude that the relation

$$\det \left[\begin{pmatrix} A_0 & -B_0 & 0, \dots \\ C_1^{(1)} & A_1 & -B_1 & 0, \dots \\ & \ddots & \ddots & \\ 0, \dots, 0 & D^{(q+1)}, D^{(q)}, \dots, & C_N^{(1)} & A_N \end{pmatrix} - EI \right] = 0, \quad N \rightarrow \infty, \quad (4.19)$$

defines the bound-state energies for the potential (4.1).

V. POTENTIALS EQUIVALENT TO POLYNOMIALS OF AN EVEN DEGREE

The pair of forces (4.1) and

$$V(r) = g_0 + g_1 r^2 + \dots + g_{2q+1} r^{4q+2}, \quad g_{2q+1} = a^2 > 0, \quad (5.1)$$

represent the complete class of the general "fractionally anharmonic" oscillators (2.1) [cf. Sec. II (d)]. From the purely formal point of view the new interaction (5.1) contains one more coupling, but we may postulate

$$\psi(r) = r^{l+1} \exp(-g(r)) \sum_{n=0}^{\infty} p_n r^{2n}, \quad (5.2)$$

$$g(r) = \sum_{j=1}^{q+1} \frac{\beta_j r^{2j}}{2j}, \quad \beta_{q+1} = a > 0,$$

and arrive again at the $(2q+2)$ -term recurrences

$$\begin{aligned} B_n p_{n+1} & = (A_n - E) p_n + \sum_{j=1}^q p_{n-j} C_n^{(j)} \\ & + \sum_{j=q+1}^{2q} p_{n-j} D^{(j-q)}, \\ n & = 0, 1, \dots, \quad p_{-1} = p_{-2} = \dots = 0, \end{aligned} \quad (5.3)$$

analogous to Eq. (4.3).

Before going into details, let us recall a close analogy between (5.1) and the harmonic oscillator [= a special case of (5.1) with $q = 0$], and reparametrize (5.1):

$$\begin{aligned} V(r) & = W_1(r) + W_2^2(r) r^2, \\ W_1(r) & = c_0 + c_1 r^2 + \dots + c_q r^{2q}, \\ W_2(r) & = \gamma_1 + \gamma_2 r^2 + \dots + \gamma_{q+1} r^{2q}, \quad \gamma_{q+1} = a. \end{aligned} \quad (5.4)$$

In terms of the new couplings, we may express then also the explicit definitions of coefficients in (5.3),

$$\begin{aligned}
B_n &= (2n+2)(2n+2l+3), \\
A_n &= (4n+2l+3)\beta_1 + c_0 \equiv C_n^{(0)} + E, \\
C_n^{(j)} &= (4n+2l+3-2j)\beta_{j+1} + c_j + d_j, \\
d_j &= \sum_{i=0}^{j-1} (\gamma_{j-i} + \beta_{j-i})(\gamma_{i+1} - \beta_{i+1}), \\
D^{(j)} &= \sum_{i=j-1}^{q-1} (\gamma_{q+j-i} + \beta_{q+j-i})(\gamma_{i+1} - \beta_{i+1}), \\
j &= 1, 2, \dots, q,
\end{aligned} \tag{5.5}$$

with the possible explicit transition between the definitions (5.1) and (5.4) of the potential,

$$\begin{aligned}
\gamma_{q-m} &= \frac{1}{2a} \left(g_{2q-m} - \sum_{i=1}^m \gamma_{q+m-i} \gamma_{q+1-i} \right), \\
m &= 0, 1, \dots, q-1,
\end{aligned} \tag{5.6}$$

etc.

In comparison with the preceding section, the change of variables (4.7) finds its present counterpart in the relations

$$p_{n+1} = \frac{\lambda^n h_n \Gamma(1 + (n+v)/(q+1))}{\Gamma(1 + (n+1)/(q+1)) \Gamma(1 + (n+l+\frac{3}{2})/(q+1))}, \tag{5.7}$$

$$4v = 2l+3-2q + (1/a)(c_q + d_q), \quad n = 0, 1, \dots, \quad \lambda^{q+1} = a/(q+1).$$

This transformation converts our basic difference equation (5.3) into the $q+1$ different equivalents

$$h_n - h_{n-q-1} = \sum_{m=0}^{q-1} (w_{q-m} h_{n+m-q} + w_{2q+1-m} h_{n+m-2q-1}), \tag{5.8}$$

where the auxiliary parameter $\lambda = \lambda_{(k)}$ has a form similar to Eq. (4.9),

$$\lambda = \lambda_{(k)} = \left| \left(\frac{a}{q+1} \right)^{1/(q+1)} \right| \exp \left(i \frac{2\pi k}{q+1} \right), \quad k = 0, 1, \dots, q. \tag{5.9}$$

The coefficients

$$\begin{aligned}
w_{q-m} &= \frac{C_n^{(q-m-1)} \Gamma(1 + (n+m-q+v)/(q+1)) \Gamma(1 + (n+1)/(q+1)) \Gamma(1 + (n+l+\frac{3}{2})/(q+1))}{B_n \lambda^{q-m} \Gamma(1 + (n+v)/(q+1)) \Gamma((n+m+2)/(q+1)) \Gamma((n+m+l+\frac{3}{2})/(q+1))}, \\
w_{2q+1-m} &= \frac{D^{(q-m)}(q+1) \Gamma((n+m-q+v)/(q+1)) \Gamma(1 + (n+1)/(q+1)) \Gamma(1 + (n+l+\frac{3}{2})/(q+1))}{a B_n \lambda^{q-m} \Gamma(1 + (n+v)/(q+1)) \Gamma((n+m+1-q)/(q+1)) \Gamma((n+m+l+\frac{3}{2}-q)/(q+1))}, \\
m &= 0, 1, \dots, q-1,
\end{aligned} \tag{5.10}$$

exhibit again the $n \gg 1$ asymptotic decrease

$$\begin{aligned}
w_{q-m} &= (q+1)^{-1} \beta_{q-m} \lambda_{(k)}^{-q} [n/(q+1)]^{-(m+1)/(q+1)} + \text{corrections}, \\
w_{2q+1-m} &= (q+1)^{-1} D^{(q-m)} [n/(q+1)]^{-(m+1)/(q+1)} / 4a \lambda_{(k)}^{-m} + \text{corrections},
\end{aligned} \tag{5.11}$$

displayed also in Fig. 1(b).

In full analogy with the preceding section, the simple $n \gg 1$ differential equation approximations to Eq. (5.8),

$$\begin{aligned}
(q+1)^2 \frac{d}{dn} \ln h_n &= \frac{\lambda_{(k)}^{-q}}{2} \left(\frac{n}{q+1} \right)^{-1/(q+1)} [\gamma_q + \beta_q + \text{corrections}], \\
\end{aligned} \tag{5.12}$$

give $q+1$ independent solutions

$$\begin{aligned}
h_n &= h_n^{(k)} \\
&= \exp \left[\frac{\beta_q + \gamma_q}{2q \lambda_{(k)}^q} \left(\frac{n}{q+1} \right)^{q/(q+1)} + \text{corrections} \right], \\
n &\gg 1, \quad k = 0, 1, \dots, q,
\end{aligned} \tag{5.13}$$

numbered by the subscript k used in (5.9).

The remaining q independent solutions of Eq. (5.8) with $D^{(q)} \neq 0$ may be found in the same way as above—only the suitable change of variables

$$\begin{aligned}
p_{n-q} &= \mu^n \tilde{h}_n / \Gamma(1 + (n+r)/q), \quad \mu^q = -(1/4aq) D^{(q)}, \\
r &= (1/4a) [(2l+3-2q)a + c_q + d_q],
\end{aligned} \tag{5.14}$$

differs from Eq. (5.7) and replaces Eq. (5.3) by the q relations

$$\begin{aligned}
\tilde{h}_n - \tilde{h}_{n-q} &= \frac{\mu}{(n/q)^{1/q}} \left(-\frac{D^{(q)}}{4a^2} \tilde{h}_{n+q+1} - \frac{\beta_q}{a} \tilde{h}_{n+1} \right. \\
&\quad \left. + \frac{D^{(q-1)}}{D^{(q)}} \tilde{h}_{n-q+1} \right) + \text{corrections}, \quad n \gg 1.
\end{aligned} \tag{5.15}$$

We arrive at the q missing and independent quasiconstant solutions

$$\tilde{h}_n^{(k)} = \exp(\text{const} \cdot \mu_{(k)} n^{(q-1)/q} + \text{corrections}),$$

$$n \gg 1, \quad k = 1, 2, \dots, q, \tag{5.16}$$

immediately. A similar procedure works for $D^{(q)} = 0$ as well.

The detailed form of constants in (5.16) is not needed.

When we compare (5.7) with (5.14), we see that the latter q components of p_{n+1} are suppressed by a huge overall factor, $\Gamma(\text{const} + n/q)/\Gamma(\text{const} + n/(q+1))$

$$\approx \Gamma(\text{const} + n/[q(q+1)]), \quad n \gg 1.$$

We may omit them completely from the general solution of Eq. (5.3) and write the latter, within the present error bounds, in the form

$$\frac{\Gamma(1 + (n+1)/(q+1))\Gamma(1 + (n+l+\frac{3}{2})/(q+1))}{\Gamma(1 + (n+v)/(q+1))} p_{n+1} = \sum_{k=0}^q c_k \lambda_{(k)}^n h_n^{(k)}, \quad n \gg 1, \quad (5.17)$$

fully analogous to our former Eq. (4.14).

In what follows we shall assume a choice of β_q such that

$$\beta_q + \gamma_q > 0. \quad (5.18)$$

As a consequence, the right-hand-side sum in (5.17) becomes dominated by the $k=0$ component [cf. Eq. (4.15)]. Indeed, for $k \neq 0$, we have always a nonzero imaginary part in the factor

$$1/\lambda_{(k)}^q = \lambda_{(0)}^{-q} \exp(-2\pi i k q/(q+1)) = \lambda_{(0)}^{-q} \exp(-2\pi i (q+1-k)/(q+1)).$$

An insertion in (5.2),

$$\psi(r) = r^{q+1} \exp(-g(r)) [O(r^{2N}) + \phi_N(r)], \quad r \gg 1,$$

$$\phi_N(r) = \sum_{n=N}^{\infty} p_n r^{2n},$$

enables us to write

$$\begin{aligned} \phi_N(r) &\approx c_0 \int_N^{\infty} dn r^{2n} \frac{\lambda_{(0)}^n}{\Gamma(1 + (n+1)/(q+1))} \\ &\times \exp O(n^{q/(q+1)}) \\ &\approx c_0 \sum_{m \geq N/(q+1)} \frac{r^{2m(q+1)}}{m!} \\ &\times \exp\left(m \ln \frac{a}{q+1} + O(m^{q/(q+1)})\right) \\ &\approx c_0 e^{a r^{2q+2}/(q+1)}, \quad r \gg 1, \end{aligned} \quad (5.19)$$

in full agreement with the unphysical asymptotic growth [(2.5) with $d_1 \neq 0$] whenever $c_0 = c_0(E) \neq 0$. Conversely, the zeros of $c_0(E)$ will coincide with the zeros of $\psi(r)$, $r \gg 1$, as well as with the zeros of the Hill determinant,

$$\det \begin{bmatrix} A_0 & -B_0 & 0 & \dots & 0 \\ C_1^{(1)} & A_1 & -B_1 & 0 & \dots \\ & \ddots & \ddots & \ddots & \\ 0, \dots, 0 & D^{(q)}, \dots, D^{(1)} & C_N^{(q)}, \dots, C_N^{(1)} & A_N \end{bmatrix} - EI$$

$$= 0, \quad N \gg 1. \quad (5.20)$$

The proof of the validity of the method [Eq. (3.4)] for the potential (5.1) is completed.

VI. SUMMARY

In this paper we have considered the complete class of the fractional potentials (1.4). We have transformed (i) the potentials into their two "canonical" forms, (4.1) and (5.1), in accord with Ref. 15; (ii) the regular wave functions $\psi(r)$, $r \in (0, \infty)$, into their appropriate Taylor coefficients p_n , $n = 1, 2, \dots$ [cf. Eqs. (4.2) and (5.2), respectively]; (iii) the radial (ordinary differential) Schrödinger equation into its difference equation equivalents [Eqs. (4.3) and (5.3), respectively]; and (iv) the standard physical boundary condition

$$\psi(r) = 0, \quad r \rightarrow \infty, \quad (6.1)$$

into its "Hill-determinant" equivalent

$$p_N = 0, \quad N \rightarrow \infty. \quad (6.2)$$

The core of the paper lies in a complete asymptotic solution of the difference equations, showing that the asymptotically dominant component of p_n is in a one-to-one correspondence to the $r \gg 1$ asymptotically dominant component of the wave function $\psi(r)$ itself. Our main result [the equivalence between the two boundary conditions (6.1) and (6.2)] is a property of our particular choice of the transformation $\psi(r) \rightarrow p_n$, and, in light of the existing counterexamples,¹¹ it need not be valid in general, of course.

¹S. Flügge, *Practical Quantum Mechanics I* (Springer, Berlin, 1971), pp. 68 and 160.

²See Ref. 1, p. 80.

³J. H. Wilkinson, *The Algebraic Eigenvalue Problem* (Clarendon, Oxford, 1965).

⁴C. A. Ginsburg, *Phys. Rev. Lett.* **48**, 839 (1982).

⁵A. Hautot, *Phys. Rev. D* **33**, 437 (1986).

⁶V. Singh, S. N. Biswas, and K. Datta, *Phys. Rev. D* **18**, 1901 (1978); *Lett. Math. Phys.* **3**, 73 (1979).

⁷See, for example, S. N. Biswas, K. Datta, R. P. Saxena, P. K. Srivastava, and V. S. Varma, *Phys. Rev. D* **4**, 3617 (1971); *J. Math. Phys.* **14**, 1190 (1973); D. P. Datta and S. Mukherjee, *J. Phys. A* **13**, 3161 (1980); D. Masson, *J. Math. Phys.* **24**, 2074 (1983).

⁸M. Znojil, *Lett. Math. Phys.* **5**, 405 (1981).

⁹A. H. Wilson, *Proc. R. Soc. London Ser. A* **118**, 617 (1928).

¹⁰M. Tater, *J. Phys. A* **20**, 2483 (1987).

¹¹R. N. Chaudhuri, *Phys. Rev. D* **31**, 2687 (1985).

¹²M. Znojil, *Phys. Rev. D* **26**, 3750 (1982).

¹³M. Znojil, *Phys. Rev. D* **34**, 1224 (1986).

¹⁴C. Itzykson and J. B. Zuber, *Quantum Field Theory* (McGraw-Hill, New York, 1980).

¹⁵M. Znojil, *J. Math. Phys.* **24**, 1136 (1983).

¹⁶R. G. Newton, *Scattering Theory of Waves and Particles* (McGraw-Hill, New York, 1982), Chap. 12.

¹⁷E. L. Ince, *Ordinary Differential Equations* (Dover, New York, 1956), Chap. 10.

¹⁸M. Znojil, *J. Phys. A* **16**, 213 (1983).